Business Process Engineering
Week # 11

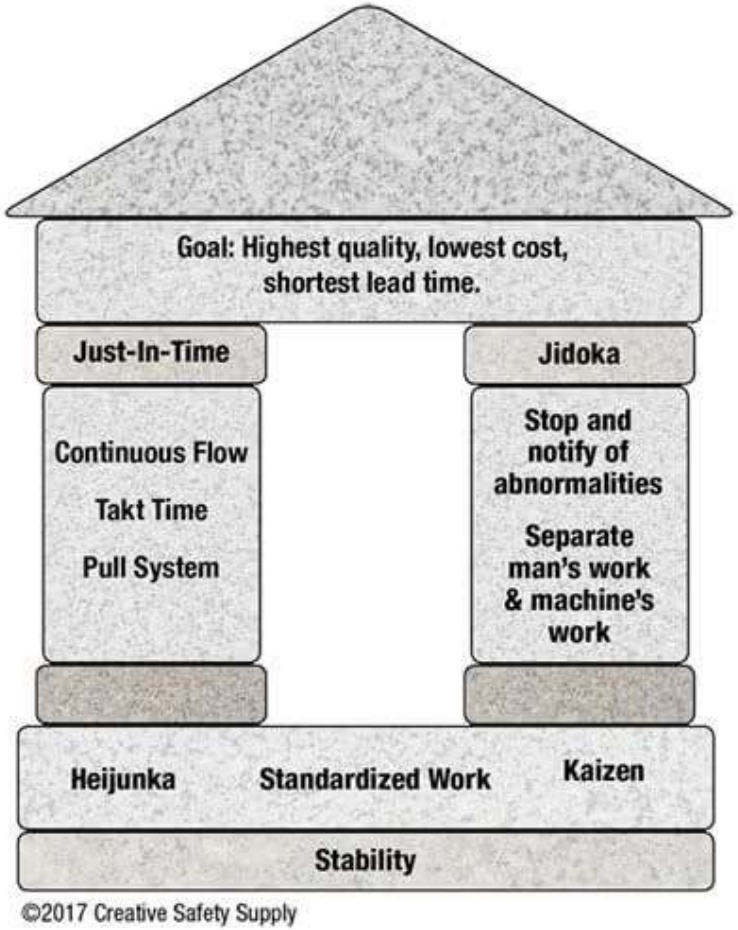
FALL 2025

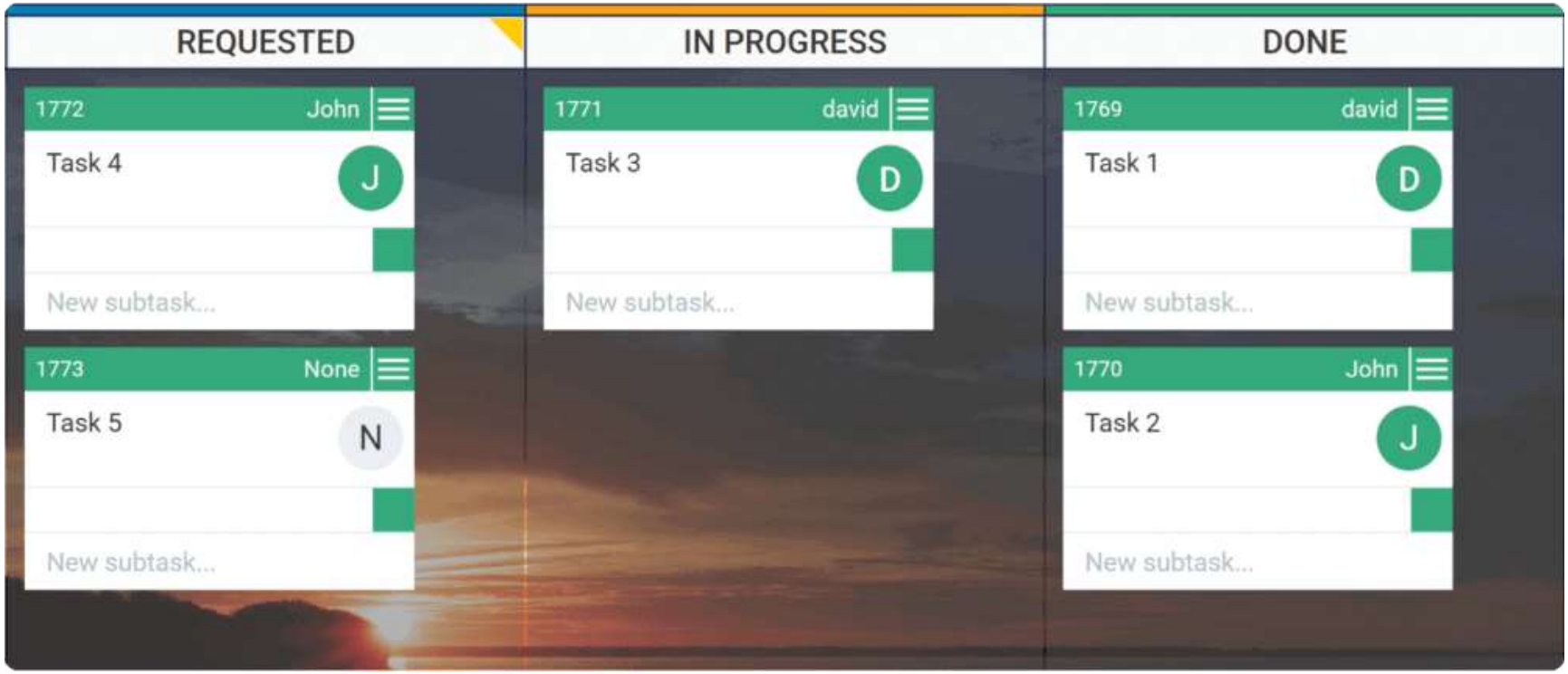
What is Kaizen?

- Kaizen (Ky'zen)
- "Kai" means "change"
- "zen" means "good (for the better)"
- Gradual, orderly, and continuous improvement
- Ongoing improvement involving everyone

History

- Modern Kaizen is based on the principles of the Toyota Production System (TPS).
- TPS is a system used in repetitive manufacturing, bu the philosophy can be applied to all operations.

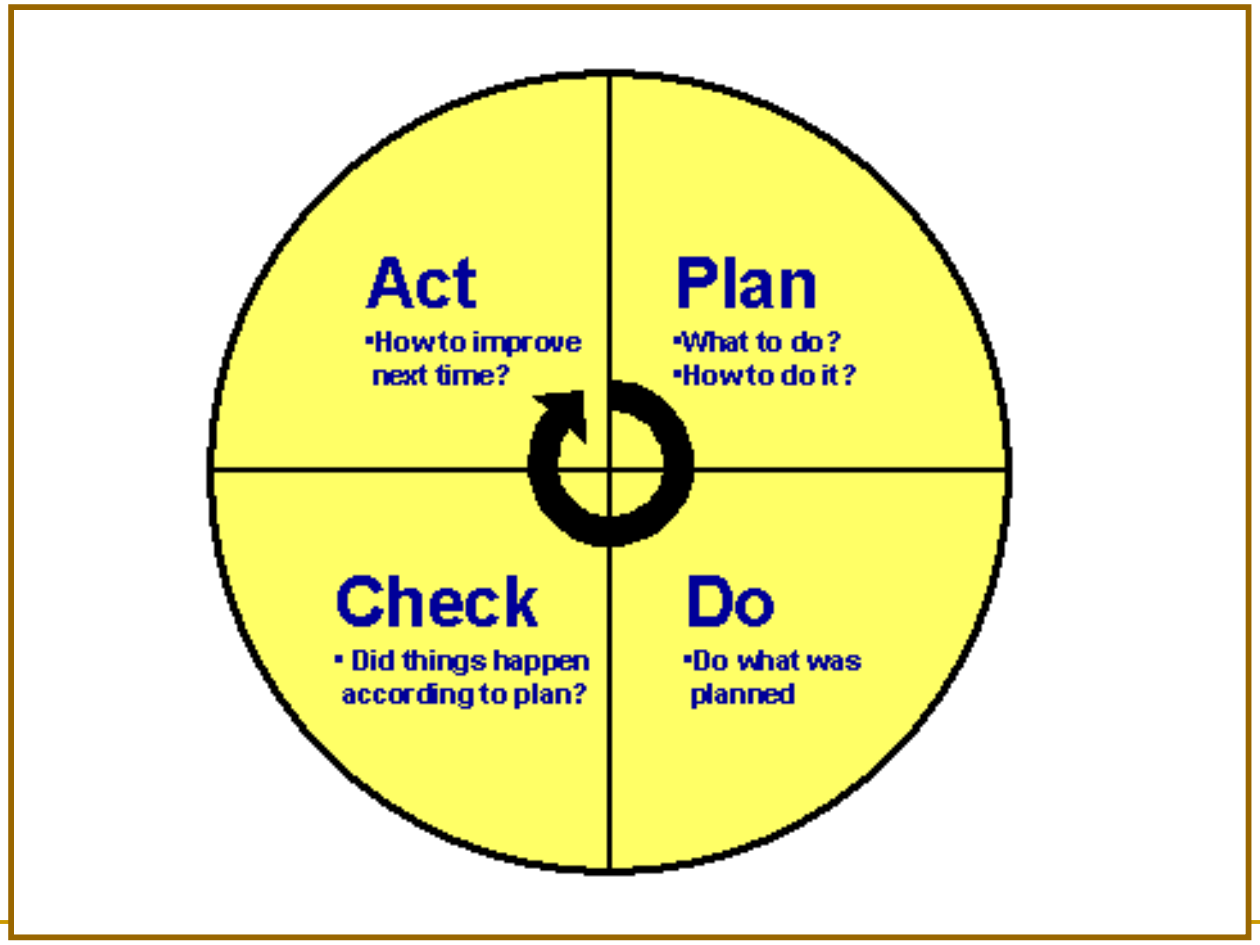



REQUESTED
IN PROGRESS
DONE
1772 John
Task 4
J
New subtask...
1773 None
Task 5
N
New subtask...
1771 david
Task 3
D
New subtask...
1769 david
Task 1
D
New subtask...
1770 John
Task 2
J
New subtask...

How to Kaizen

- Identify the customer
- Deming Cycle
  - Plan – identify what to change and how to do it
    - Current state
    - Future state
    - Implementation plan
  - Do – execute the improvement
  - Check – ensure the improvement works
  - Act – future and ongoing improvements
  - Repeat

Deming Cycle



Identify the Customer

- Value added is always determined from the customer's perspective.
- Who is the customer?
- Every process should be focused on adding value to the customer.
- Anything that does not add value is waste.
- Some non-valued added activity is necessary waste ("NVA-R")
  - Regulatory
  - Legal

Types of Waste

- Overproduction
- Excess inventory
- Defects
- Non-value added processing
- Waiting
- Underutilized people
- Excess motion
- Transportation

Identify the Current State

- Crucial first step in process improvement
- Deep understanding of the existing processes and dependencies
- Identify all the activities currently involved in developing a new product
- Observe the process first hand
- Identify Value Added (VA), Non-Value Added Required (NVA-R), and Non-Value Added (NVA)
- Generally creates more questions than answers

Identify the Current State



Identify VA, NVA-R, and NVA



Tools

- Flow Charts
- Cause and Effect Diagrams
- Check Sheets
- Histograms
- Pareto Charts
- Scatter Diagram
- Control Charts

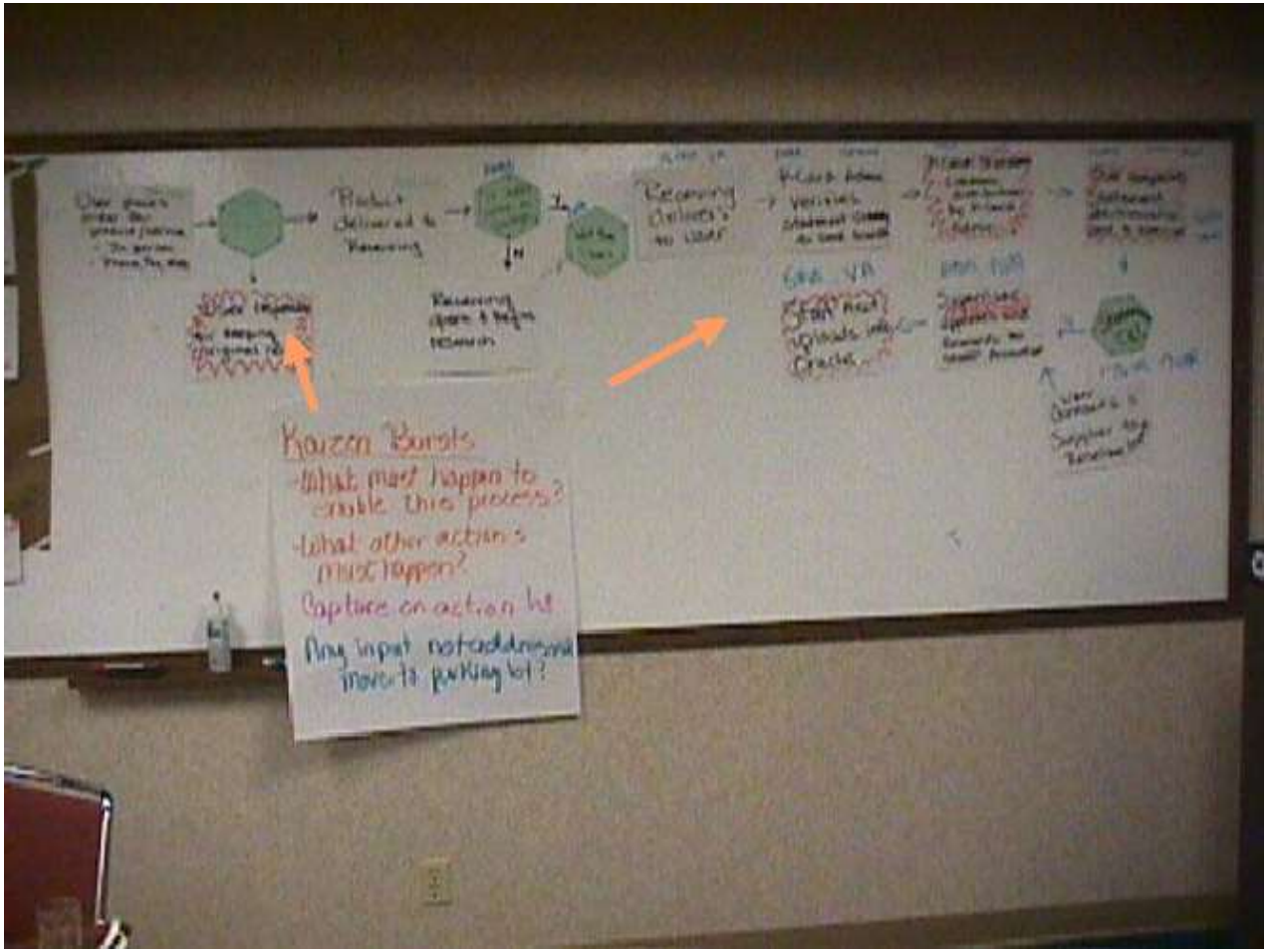
Cause and Effect Diagram (CED)

- Fishbone diagram
- Relates causes to effects
- Benefits?
- Disadvantages?
- How?
  - Start with effect and work back to possible causes

Brainstorm and Analyze

- Kaizen team brainstorming to develop new process
- Post improvement ideas on map or by category
  - Workflow
  - Technology
  - People / Organization
  - Procedures
- Develop detailed future state map
  - New workflow
  - Value Add and Non-Value Add
  - Cycle times
  - Identify Kaizen “bursts” (immediate radical change)

Develop the Future State



Implementation Plan

- Think global / systems optimization
- Maximum impact to process
- Speed of implementation – create small victories
- Cost-benefit analysis

Implementation Plan

- Will new skill sets be required, and how to achieve them?
- Is the current organization structure sufficient?
- Are there cultural issues?
- Is there potential for “push back”?
- Any implications for suppliers?
- Implications for customers?
- Implications for team members?
- Do current technologies support the new process?
- Are they available and cost justifiable?
- Technology is an enabler, not a solution.
- Does the reward system support the new process?

Execute

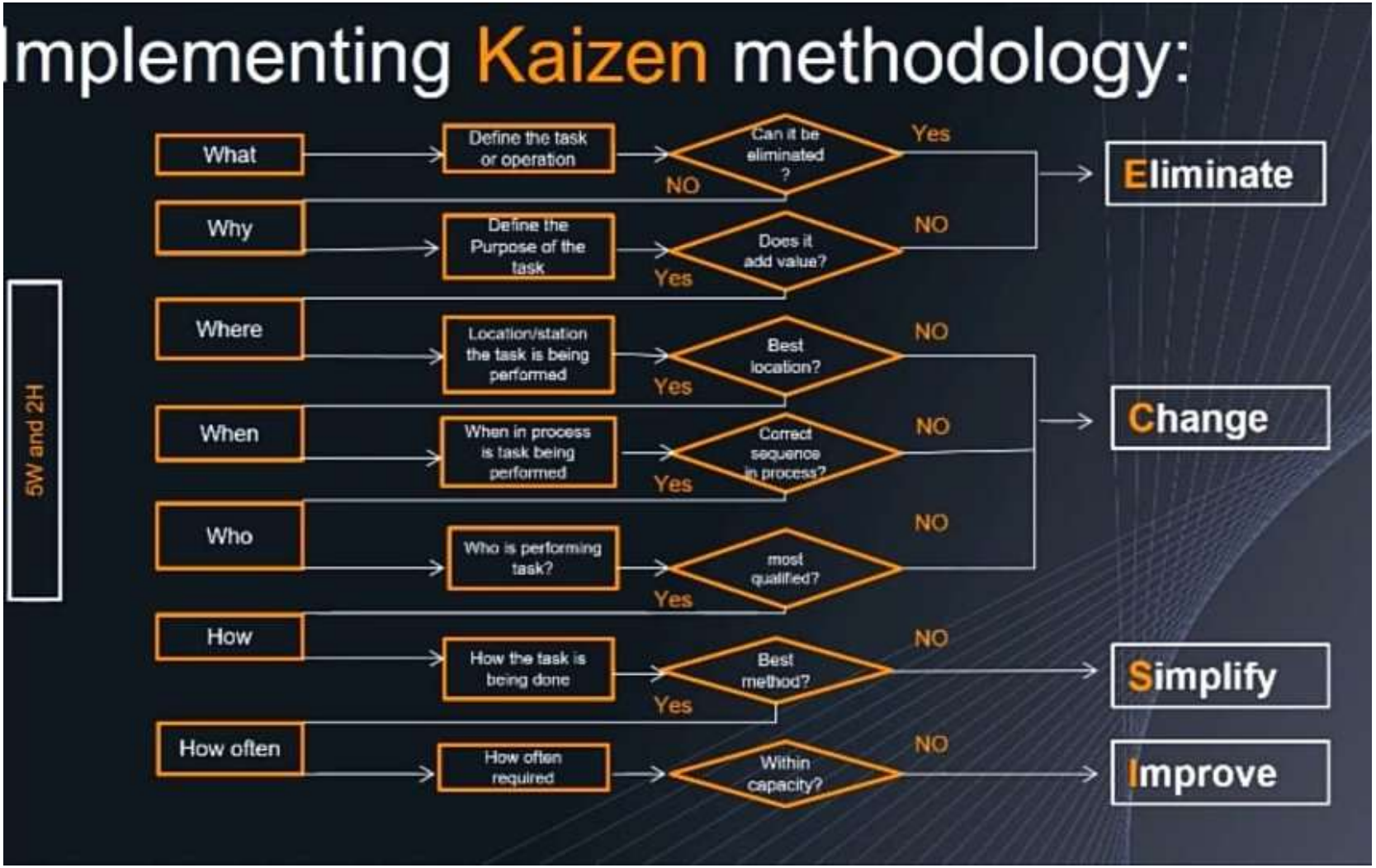
- Develop a concise, achievable milestone plan
- Communicate the plan to everyone
  - Suppliers
  - Team members
  - Customers
- Track activities in public
- Celebrate small victories and publicly analyze failures

Check and Sustain

- Meet regularly (weekly?) to review status of open implementation items
- Re-evaluate Future State regularly (quarterly?) for additional improvement
- Track results on a public Kaizen Board

Typical Results

- 40 – 60% reduction of lean time
- 10 – 15% productivity improvement
- 10 – 20% reduction in rework
- Improved communication between functions and departments
- Clearly defined customer needs throughout the value stream
- Improved customer satisfaction


Implementing Kaizen methodology:
5W and 2H
What
Define the task or operation
Can it be eliminated ?
Yes
NO
Eliminate
Why
Define the Purpose of the task
Does it add value?
NO
Yes
Where
Location/station the task is being performed
Best location?
NO
Yes
When
When in process is task being performed
Correct sequence in process?
NO
Yes
Who
Who is performing task?
most qualified?
NO
Yes
Change
How
How the task is being done
Best method?
NO
Yes
Simplify
How often
How often required
Within capacity?
NO
Improve

Kaizen Blitz

- Total focus on a defined process to create radical improvement in a short period of time
- Dramatic improvements in productivity, quality, delivery, lead-time, set-up time, space utilization, work in process, workplace organization
- Typically five days (one week) long

Kaizen Blitz - Agenda

- Day 1: Setting the scene
  - Meet the team, training
- Day 2: Observe the current process
  - Flowchart, identify waste, identify root causes
- Day 3: Develop the future state process
  - Brainstorm and flowchart (typically the longest day!)
- Day 4: Implement the new process
  - Plan, communicate, implement, modify
- Day 5: Report and analyze
  - Performance vs expectations

Typical Kaizen Weekly Schedule

	Day 1 – Orientation	Day 2 – Understand current situation	Day 3 – Develop future state design	Day 4 – Make the improvements	Day 5 – Report & celebrate
Morning	• Welcome & Introductions • Set expectations • Provide focused training • Identify the Customer • Define Kaizen Charter	• Analyze current state • Brainstorm ideas • Root cause analysis • Prioritize solutions • Future state design	• Future state design • Mid-week review • Implement improvements	• Implement improvements • Try out or simulate new process • Train participants • Tweak design as needed	• Finalize future state • Complete training on standard work • Final presentation
Lunch					• Lunch & celebration
Afternoon	• Document current state – process map or value stream map, identify waste (opportunities), data collection, etc. • Establish current state • Day 1 summary • Set objective(s) for Day 2	• Analyze current state • Brainstorm ideas • Root cause analysis • Prioritize solutions • Future state design • Day 2 summary • Set objective(s) for Day 3	• Implement improvements • Try out or simulate new process • Train participants • Tweak design as needed • Day 3 summary • Set objective(s) for Day 4	• Create and implement standard work • Measure future state • Calculate actual benefits • Day 4 summary • Set objective(s) for Day 5	

Kaizen Blitz Rules

- Be open to change
- Stay positive
- Speak out if you disagree
- See waste as an opportunity
- No blame environment
- Treat others as you want to be treated
- Ask the silly questions, challenge the givens
- Creativity before capital
- Understand the data and principles
- Just do it!

Roadblocks

- Too busy to study it
- A good idea but the timing is premature
- Not in the budget
- Theory is different from practice
- Isn't there something else for you to do?
- Doesn't match corporate policy
- Not our business – let someone else analyze it
- It's not improvement – it's common sense
- I know the result even if we don't do it
- Fear of accountability
- Isn't there an even better way?